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# Does a decade of the rally-point scoring system impact the characteristics of elite badminton matches?

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## ABSTRACT

This study aimed to investigate game characteristics in men's singles badminton matches at world championships under a decade of rally-point scoring system. Time interval has been hypothesised to affect timing factors, such as game duration, rally time, rest time, rest at point 11, and rest time between games. Overall, 56 matches ( $n = 127$  games) were analysed using video footage. We found that year affected all variables from 2006 to 2017, except for the total points played. Consequently, game duration, rally time, rest time, rest time at point 11 and rest time between games increased by 54.0% ( $p = 0.002$ ,  $r = 0.407$ ), 62.2% ( $p = 0.000$ ), 49.3% ( $p = 0.000$ ), 44% ( $p < 0.001$ ,  $r = 0.36$ ) and 74.9% ( $p = 0.000$ ,  $r = 0.50$ ), respectively. Furthermore, we suggest that game characteristics in men's singles badminton matches evolve towards longer rallies with longer rest time, rest time at point 11 and rest time between games, exceeding the limits of the badminton regulation. Therefore, stricter rule enforcement, the introduction of a time limit for the rest time between the rallies, which is similar to the shot clock implemented in tennis for the service execution or a change of the scoring system appears to be necessary to ensure a further reduction of the game duration.

## ARTICLE HISTORY

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## KEYWORDS

Badminton; match analysis; elite performers; world championships; rally-point scoring system

## 1. Introduction

Badminton is a sport with more than 330 million active participants worldwide (Badminton World Federation, 2019). The official badminton competition comprises five events as follows: men's singles (MS), women's singles (WS), men's doubles (MD), women's doubles (WD) and mixed doubles (MX) (Badminton World Federation, 2023a).

While other sports collect a large amount of match data automatically through camera systems, such as big data in soccer (Rein & Memmert, 2016), match analysis in tennis (Born & Vogt, 2021) and commercial providers, for example, for tennis Wimbledon Information System, presented by IBM (Grambow et al., 2020), badminton sports do not yet have comprehensive match data collection. Notably, the first step was the introduction of Hawk-Eye technology in 2014; however, this was only in some elite events (Primo et al., 2019).

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Therefore, only a few studies have reported differences in game characteristics between events (Abian-Vicen et al., 2013; Abián-Vicén et al., 2018; Alcock & Cable, 2009; Gawin et al., 2015; Torres-Luque et al., 2020) with a main research focus on MS (Abdullahi & Coetzee, 2017; Abián et al., 2014; Chiminazzo et al., 2018; Laffaye et al., 2015).

In 2006, the Badminton World Federation (BWF) launched a novel “rally-point scoring” system with a best-of-three game, up to 21 points, to win the game with at least two points difference and up to a maximum of 30 points within a set. Until then, points could only be won on own serve, which meant that the maximum number of rallies in a match was not limited. With the new scoring system, every rally, regardless of the server, leads to a point being awarded, which limits the maximum number of rallies in a match. This system was designed to shorten the duration of badminton matches to make them more exciting to watch, attract more sponsors, expand media exposure, and eventually increase the sport’s popularity (Chen & Chen, 2008). A few studies have analysed match characteristics by comparing the old scoring system (the best-of-three games of 15 points for doubles and MS and 11 points for WS) with the rally-point scoring system (Chen & Chen, 2008; Chen et al., 2011; Ming et al., 2008). However, the differences between the two scoring systems have been discussed. Chen et al. (2011) reported decreased match duration, work density, rest time, effective playing time, and the number of rallies in the rally-point scoring system.

Currently, badminton has been played for over 15 years under the rally-point scoring system. Only a few studies have analysed the evolution of badminton matches within the current scoring system; comparing two Olympic Games (Abián et al., 2014; Laffaye et al., 2015; Torres-Luque et al., 2020), the evolution within an Olympic Games tournament (Chiminazzo et al., 2018) or between two Olympic Games qualification periods (Iizuka et al., 2017). Abián et al. (2014) analysed MS at the 2008 and 2012 Olympic Games and found that game duration, real-time played, rally time, shots per rally, rest time at 11, and rest time between games were longer in London 2012 than they were in Beijing 2008. They concluded that badminton is evolving towards longer rallies with greater rest intervals, which result in longer matches. Laffaye et al. (2015) compared the MS finals at the Olympic Games from 1992 to 2012 and found differences in rally time, rest time, number of shots per rally, and an increase in frequencies by 34%. Iizuka et al. (2017) investigated matches from 12 Superseries tournaments during the 2008 Beijing and 2016 Rio de Janeiro Olympic Games qualification periods. They reported that the durations of the matches for tournaments during the 2015 qualification period increased compared to the 2007 qualification period. A more detailed comparison of match characteristics between the Japan Open Superseries from 2007 to 2015 shows that rally time in MS and WS events as well as rest time in all events, was longer in 2015. Iizuka et al. (2017) suggested that the increased rest time and rally time are associated with a longer match duration during the 2016 Rio de Janeiro Olympic Games qualification period. Chiminazzo et al. (2018) found differences between the group and play-off stages in the 2016 Rio de Janeiro Olympics, concluding that play-off matches were more intense than matches in the group phase (i.e. match duration, total rest time, points played, and shots per rally were all higher than in group phase matches). Another study, which examined the 2012 and 2016 Olympic MS, reported no difference in match duration in MS but an increase in the duration of the third game from 2012 to 2016 (Torres-Luque et al., 2020).

Previous studies analysing the Badminton world class used the Olympic Games, held only every four years, as data set and found different developments in terms of match and game duration (Abián et al., 2014; Laffaye et al., 2015; Torres-Luque et al., 2020). Since different climatic and geographical conditions (e.g. London 2012 and Rio 2016) could have had an impact on results as well as analysing a development by comparing only two years, a long-term approach, analysing a world-class tournament in consecutive years could be beneficial and minimise the above stated risks. However, a comprehensive long-time approach covering more than a decade of the rally-point scoring system yearly using the highest tournament, e.g. the World Championships that took place every year without the year the Olympic Games, is currently lacking.

In 2014, the BWF further reviewed the scoring system, discussing it at its annual member's forum to eventually advance badminton (Badminton World Federation, 2014). In the same year, BWF decided to test an experimental scoring system with a best-of-five game of up to 11 points (Høyer, 2014), instated in the BWF laws of badminton (Badminton World Federation, 2015) as an alternative scoring system. Even tournaments have been held under the experimental scoring system (Høyer, 2016) with feedback collected from the involved participants (i.e. players, trainers and spectators) (Badzine, 2014). Yang et al. (2016) compared the experimental scoring system to the current scoring system following a model-based probability of the server winning the rally and reported a reduction in single-match duration when using the experimental scoring system; however, this did not affect the match outcome when using the current scoring system.

In 2018, the BWF council proposed the introduction of a novel and enhanced scoring system based on a best-of-five game system of up to 11 points, further aiming to increase the excitement of the sport at the elite level and to shorten the length of matches (Badminton World Federation, 2018). However, this failed member's majority voted to retain the best-of-three-games system of up to 21 points as the official scoring system (Høyer, 2018). In 2019, BWF president Paul Erik Hoyer renewed the idea of changing the scoring system after the 2020 Olympic Games (Badzine, 2019). To the best of our knowledge, this is the first approach that aimed to cover a decade of badminton matches using the rally-point scoring system based on the almost annually played World Championships to provide insights into the hypothesised changes in timing characteristics of badminton matches. The findings may be of greater value for an ongoing discussion to change the scoring system.

Therefore, this study aimed to investigate game characteristics in MS elite badminton over a decade (2006–2017) under the rally-point scoring system. The following was hypothesised: an increase in game duration can be observed over the decade, longer rally time and rest time impact game duration and rest time at point 11 and rest time between games increase with an impact on the game duration over the period.

## **2. Materials and methods**

### **2.1. Participants and general procedure**

In total, 56 MS matches, which commenced at the 2006 World Championship in Madrid (Spain) until the 2017 World Championships in Glasgow (Scotland), were included in the analyses ( $n = 127$  games). Table 1 shows the distribution. Because of the selected period, all tournaments were played using the current rally-point scoring system. World

Championship tournaments are defined as the highest category (Level 1) of BWF tournaments (Badminton World Federation, 2023a), which take place annually, except in years with the Olympic Games. Competitors can be referred to as the best players in the world at the time of the tournament, given the category and frequency of the tournament, with comparable data for each year the World Championships were held. Seven matches per tournament were selected if available (not possible for all years, e.g. due to online availability and country restrictions), preferably including the quarterfinals, semi-finals, and finals.

Previous studies found that games served as the basis for analyses, providing the most appropriate structure to standardise the measurements without being limited by games in the match (Abián et al., 2014; Abian-Vicen et al., 2013).

## 2.2. Material

Video footage of matches that were played at the World Championships from 2006 to 2017 was used via open-access YouTube channels (e.g. BWF BadmintonWorld.TV archives). In the case of restricted access to a match (e.g. the match has not been stored online or has no country-specific access rights), we could obtain video footage of the respective matches from BWF directly. However, video files had been stored but not in all cases; therefore, they remained unavailable for analysis (see Table 1). Available video footage was used to perform the match analyses. Reproduction was carried out using YouTube and VLC Media Player software and the timing factors were measured using this software's tools. One analyst, who received extensive training in the methods and procedures of this study, performed the observations. The analyst was instructed to carefully observe all videos and record predefined parameters (see Section 2.3) for statistical analyses.

## 2.3. Variables

The independent variables were the year and place where the matches were played (Madrid [Spain] 2006; Kuala Lumpur [Malaysia] 2007; Hyderabad [India] 2009; Paris [France] 2010; London [England] 2011; Guangzhou [China] 2013; Copenhagen [Denmark] 2014; Jakarta [Indonesia] 2015 and Glasgow [Scotland] 2017).

The dependent variables were the timing factors which include game duration (the time that elapsed from the first service until the shuttlecock touched the ground on the

**Table 1.** Number of matches and games analysed at the World Championships.

| Year          | Location                | Rounds                     | Matches   | Games      |
|---------------|-------------------------|----------------------------|-----------|------------|
| 2006          | Madrid [Spain]          | R32, QF, QF, SF, SF, F     | 6         | 15         |
| 2007          | Kuala Lumpur [Malaysia] | QF, QF, SF, F              | 4         | 9          |
| 2009          | Hyderabad [India]       | QF, SF, SF, F              | 4         | 9          |
| 2010          | Paris [France]          | R32, QF, QF, QF, SF, SF, F | 7         | 16         |
| 2011          | London [England]        | R32, QF, QF, QF, SF, SF, F | 7         | 17         |
| 2013          | Guangzhou [China]       | R32, QF, QF, QF, SF, SF, F | 7         | 17         |
| 2014          | Copenhagen [Denmark]    | QF, QF, QF, QF, SF, SF, F  | 7         | 15         |
| 2015          | Jakarta [Indonesia]     | R32, QF, QF, QF, SF, SF, F | 7         | 14         |
| 2017          | Glasgow [Scotland]      | QF, QF, QF, QF, SF, SF, F  | 7         | 15         |
| <b>Total:</b> |                         |                            | <b>56</b> | <b>127</b> |

last point of game, including rest time periods between points and time point 11); rally time (the time from the service until the shuttlecock touched the ground on each point); rest time (the time that elapsed from when the shuttlecock touched the ground until the next service was performed); rest time at point 11 (when any player scored the 11th point for the first time in the game, the time that elapsed from when the shuttlecock touched the ground until the next serve); rest time between games (when any player wins the game, the time that elapsed from the shuttlecock touched the ground until the first service of the following game) and work density (rally time divided by rest time) and the following notational factors: total points played (total number of points played by both players); shots per rally (total number of times the shuttle was hit by both players from the serve until it hit the ground) and shot frequency (shots per rally divided by rally time).

We selected the variables listed below to comprise an effective evaluation index for analysing the timing factors and the notational structure of badminton matches based on previous studies (Abián et al., 2014; Cabello Manrique, 2003; Faude et al., 2007; Abián et al., 2014; Abian-Vicen et al., 2013). The reliability of the timing factors was assessed using Hopkins's (2000) intraclass correlation coefficient (ICC). Based on inter-operator agreement, the ICCs were found to be high for all the variables (from 0.99 to 1.00 with typical errors expressed as a coefficient of variation from 0 to 0.9 %; Koo & Li, 2016).

## 2.4. Statistical analysis

A Microsoft Excel spreadsheet (Microsoft, Germany) was used for data collection. SPSS v. 29.0 (SPSS Inc., U.S.A.) was used to perform descriptive and inferential statistical tests and to compute means, standard deviations and ranges.

All variables were evaluated for normal distribution using the Shapiro – Wilk test. If applicable, an analysis of variance (ANOVA) was used to establish the differences in normally distributed variables between the nine World Championship tournaments. Bonferroni corrections were applied for post-hoc analyses. Effect sizes were estimated according to Cohen (1988) using partial eta squares ( $\eta^2$ ), indicating a small ( $\eta^2 > 0.01$ ), medium ( $\eta^2 > 0.06$ ) or large ( $\eta^2 > 0.14$ ) effect.

If the normal distribution was violated (rest time at 11 points or rest time between games), Kruskal – Wallis comparisons were calculated for the factor “year”. Dunn – Bonferroni corrections were applied for post-hoc analysis. The effect size was estimated according to Cohen (1992) by  $r$ -value, indicating a small ( $r = 0.10$ ), medium ( $r = 0.30$ ), or large ( $r = 0.50$ ) effect.

Statistical significance was set at  $P < 0.05$ . Data in the text and tables are presented as the mean  $\pm$  standard deviation (SD), and the figures show the mean  $\pm$  standard error (SE).

## 3. Results

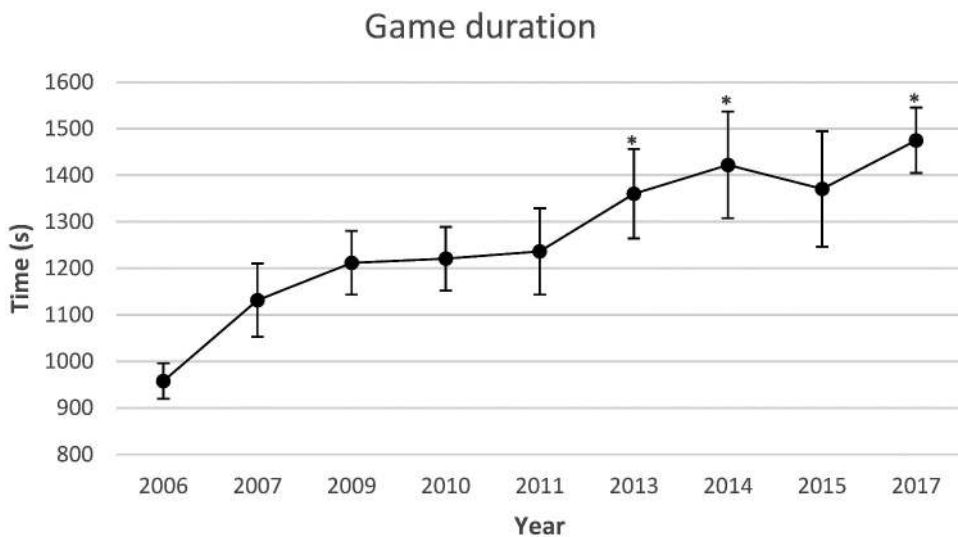
Table 2 presents an overview of the developments of timing and notational factors and the results of the Anova and Kruskal-Wallis test. Notably, the year had a significant effect on the game duration ( $f = 0.467$ ), rally time ( $f = 0.619$ ), rest time ( $f = 0.634$ ), shots per rally ( $f = 0.602$ ), shot frequency ( $f = 0.537$ ), work density ( $f = 0.433$ ) and shots per game ( $f = 0.511$ ). The effect size corresponded to a strong effect, according to Cohen (1988), for

Table 2. Mean  $\pm$  standard deviation and ANOVA or Kruskal-Wallis results obtained in each timing factor.

| Games                                   | 2006<br>$n = 15$  | 2007<br>$n = 9$    | 2009<br>$n = 9$    | 2010<br>$n = 16$   | 2011<br>$n = 17$   | 2013<br>$n = 17$   | 2014<br>$n = 15$   | 2015<br>$n = 14$   | 2017<br>$n = 15$   | F        | p     | $\eta_p^2$ |
|-----------------------------------------|-------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|----------|-------|------------|
| Game duration (s)                       | 957.6 $\pm$ 147.8 | 1131.6 $\pm$ 236.1 | 1211.9 $\pm$ 205.3 | 1220.4 $\pm$ 273.1 | 1236.1 $\pm$ 382.2 | 1360.3 $\pm$ 395.4 | 1422.3 $\pm$ 443.1 | 1370.4 $\pm$ 463.6 | 1474.9 $\pm$ 272.8 | 8, 117   | 0.002 | 0.179      |
| Rally time (s)                          | 7.4 $\pm$ 1.6     | 8.9 $\pm$ 1.2      | 9.7 $\pm$ 1.2      | 9.8 $\pm$ 2.1      | 10.4 $\pm$ 2.4     | 11.5 $\pm$ 3.0     | 10.9 $\pm$ 2.6     | 9.0 $\pm$ 1.9      | 12.0 $\pm$ 2.8     | 8, 117   | 0.000 | 0.277      |
| Rest time (s)                           | 20.3 $\pm$ 1.5    | 24.3 $\pm$ 3.6     | 25.1 $\pm$ 2.3     | 26.6 $\pm$ 4.0     | 25.7 $\pm$ 4.6     | 27.7 $\pm$ 6.5     | 30.2 $\pm$ 6.4     | 29.4 $\pm$ 7.5     | 30.3 $\pm$ 4.1     | 8, 117   | 0.000 | 0.289      |
| Shots per rally                         | 8.3 $\pm$ 1.6     | 9.5 $\pm$ 1.5      | 10.8 $\pm$ 1.1     | 10.5 $\pm$ 2.1     | 11.2 $\pm$ 2.7     | 12.5 $\pm$ 3.1     | 12.3 $\pm$ 2.8     | 10.2 $\pm$ 2.2     | 12.8 $\pm$ 2.7     | 8, 117   | 0.000 | 0.266      |
| Shot frequency (shots s <sup>-1</sup> ) | 1.15 $\pm$ 0.05   | 1.11 $\pm$ 0.05    | 1.14 $\pm$ 0.04    | 1.11 $\pm$ 0.05    | 1.11 $\pm$ 0.04    | 1.13 $\pm$ 0.06    | 1.15 $\pm$ 0.06    | 1.184 $\pm$ 0.03   | 1.10 $\pm$ 0.05    | 8, 117   | 0.000 | 0.224      |
| Work density                            | 0.38 $\pm$ 0.07   | 0.38 $\pm$ 0.04    | 0.408 $\pm$ 0.06   | 0.38 $\pm$ 0.06    | 0.42 $\pm$ 0.05    | 0.43 $\pm$ 0.11    | 0.38 $\pm$ 0.08    | 0.33 $\pm$ 0.07    | 0.41 $\pm$ 0.06    | 8, 116   | 0.008 | 0.158      |
| Total of played points                  | 37.7 $\pm$ 3.6    | 36.3 $\pm$ 4.2     | 36.0 $\pm$ 4.1     | 34.8 $\pm$ 3.5     | 35.2 $\pm$ 5.8     | 35.8 $\pm$ 4.8     | 35.8 $\pm$ 4.6     | 37.4 $\pm$ 7.2     | 36.9 $\pm$ 3.5     | 8, 117   | 0.743 |            |
| Shots per game                          | 280.1 $\pm$ 69.2  | 318.1 $\pm$ 70.5   | 385.4 $\pm$ 57.2   | 354.1 $\pm$ 91.6   | 384.8 $\pm$ 142.1  | 444.1 $\pm$ 145.0  | 419.1 $\pm$ 137.2  | 356.5 $\pm$ 90.4   | 452.9 $\pm$ 103.9  | 8, 117   | 0.000 | 0.207      |
| Rest time at point 11 (s)               | 56.8 $\pm$ 11.7   | 70.5 $\pm$ 6.5     | 66.4 $\pm$ 18.2    | 78.6 $\pm$ 7.3     | 76.8 $\pm$ 8.7     | 75.4 $\pm$ 11.6    | 69.9 $\pm$ 24.5    | 79.0 $\pm$ 11.8    | 81.9 $\pm$ 27.8    | $\chi^2$ | p     | df         |
| Rest time between games (s)             | 84.8 $\pm$ 15.4   | 98.0 $\pm$ 23.6    | 112.2 $\pm$ 44.6   | 135.8 $\pm$ 27.0   | 140.9 $\pm$ 15.2   | 143.3 $\pm$ 16.2   | 123.3 $\pm$ 42.3   | 107.0 $\pm$ 51.3   | 148.3 $\pm$ 14.7   | 33.299   | 0.000 | 8          |
|                                         |                   |                    |                    |                    |                    |                    |                    |                    |                    | 30.683   | 0.000 | 8          |

**Table 3.** Post-hoc differences with asterisks (\* for  $P < 0.05$ ) marking significances.

|                         |             | 2006   | 2007   | 2009   | 2010   | 2011   | 2013   | 2014   | 2015   | 2017   |
|-------------------------|-------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| game duration           | <b>2006</b> |        | 1.000  | 1.000  | 1.000  | 0.808  | 0.040* | 0.010* | 0.051  | 0.002* |
| rally time              | <b>2006</b> |        | 1.000  | 0.757  | 0.189  | 0.015* | 0.000* | 0.002* | 1.000  | 0.000  |
|                         | <b>2015</b> | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  | 0.127  | 1.000  |        | 0.023* |
| rest time               | <b>2006</b> |        | 1.000  | 0.922  | 0.024* | 0.110  | 0.003* | 0.000* | 0.000* | 0.000* |
| rest time at point 11   | <b>2006</b> |        | 0.047* | 0.101  | 0.000* | 0.000* | 0.000* | 0.000* | 0.000* | 0.000* |
|                         | <b>2007</b> | 0.047* |        | 0.711  | 0.047* | 0.131  | 0.309  | 0.300  | 0.107  | 0.167  |
|                         | <b>2009</b> | 0.101  | 0.711  |        | 0.013* | 0.045* | 0.135  | 0.135  | 0.037* | 0.062  |
| rest time between games | <b>2006</b> |        | 0.757  | 0.171  | 0.000* | 0.000* | 0.000* | 0.007* | 0.065  | 0.000* |
|                         | <b>2007</b> | 0.757  |        | 0.350  | 0.009* | 0.012* | 0.006* | 0.048* | 0.197  | 0.001* |
|                         | <b>2009</b> | 0.101  | 0.711  |        | 0.122  | 0.155  | 0.093  | 0.346  | 0.778  | 0.024* |
|                         | <b>2015</b> | 0.000* | 0.107  | 0.037* | 0.166  | 0.213  | 0.125  | 0.472  |        | 0.031* |
| work density            | <b>2015</b> | 1.000  | 1.000  | 0.602  | 1.000  | 0.034* | 0.003* | 1.000  |        | 0.064  |
| shots per rally         | <b>2006</b> |        | 1.000  | 0.629  | 0.477  | 0.039* | 0.000* | 0.000* | 1.000  | 0.000* |
| shot frequency          | <b>2015</b> | 1.000  | 0.028* | 1.000  | 0.003* | 0.009* | 0.068  | 1.000  |        | 0.000* |

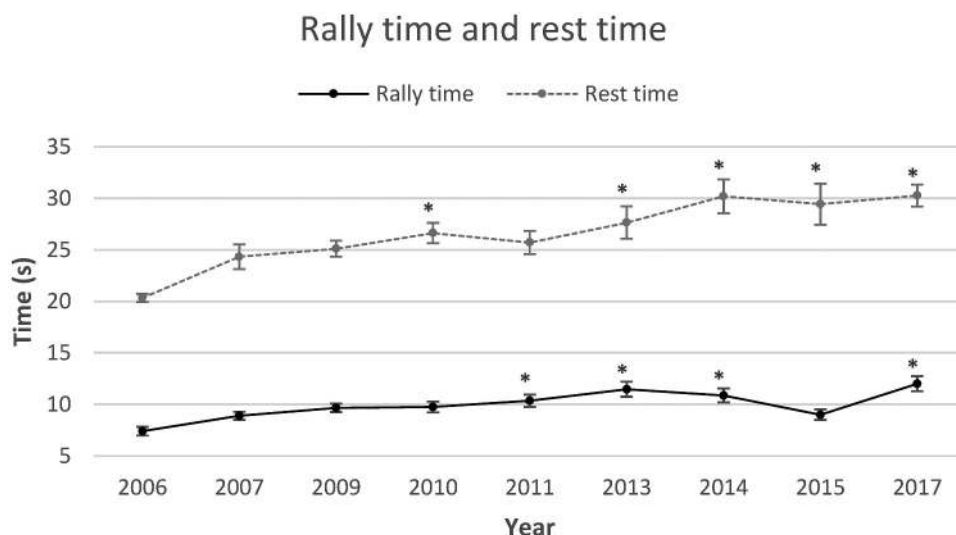
**Figure 1.** Development of game duration over the years with asterisks (\* for  $P < 0.05$ ) marking significances compared to 2006.

the parameters mentioned. However, no significant differences were found in the total number of points played.

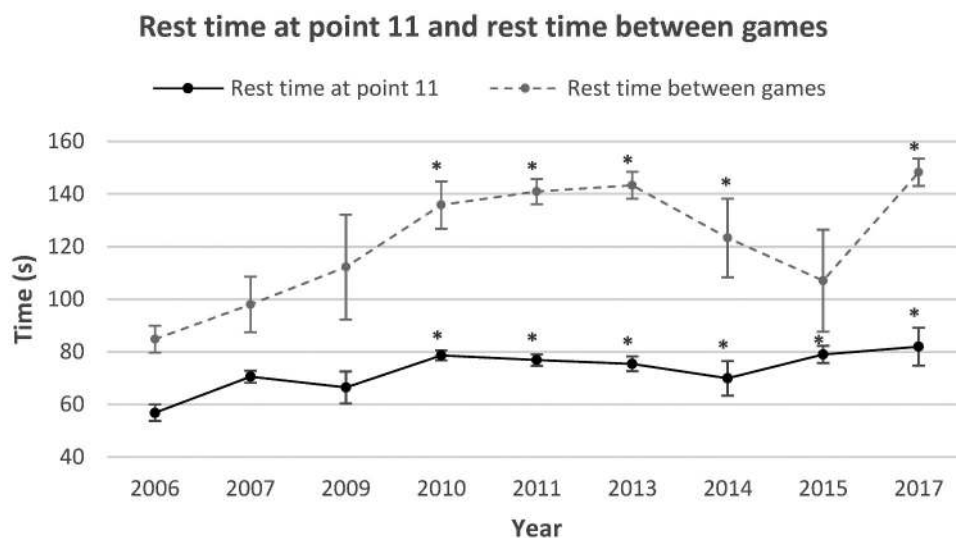
Post-hoc tests with Bonferroni correction did not show changes for every year, see [Table 3](#). However, game duration increased from 2006 to 2013 by 42.1% ( $p = 0.04$ ), to 2014 by 48.5% ( $p = 0.01$ ) and to 2017 by 54% ( $p = 0.002$ ), see [Figure 1](#). Rally time was increased in 2011, 2013, 2014 and 2017 by 40.5% ( $p = 0.015$ ), 55.4% ( $p = 0.000$ ), 47.3% ( $p = 0.002$ ) and 62.2% ( $p = 0.000$ ), respectively, compared with 2006. Furthermore, rally time in 2017 increased by 33.3% compared with 2015 ( $p = 0.023$ ). Compared with 2006, rest time increased in 2010, 2013, 2014, 2015 and 2017 by 31% ( $p = 0.024$ ), 36.5% ( $p = 0.003$ ), 48.8% ( $p = 0.000$ ), 44.8% ( $p = 0.000$ ) and 49.3% ( $p = 0.000$ ), respectively, see [Figure 2](#).

The Kruskal – Wallis test showed that the year affected rest time at 11 points (Chi-square (8) = 33.299,  $p = 0.000$ ). Subsequent post-hoc tests (Dunn – Bonferroni tests)





**Figure 2.** Development of rally time and rest time over the years with asterisks (\* for  $P < 0.05$ ) marking significances compared to 2006.).



**Figure 3.** Development of rest time at point 11 and rest time between games over the years with asterisks (\* for  $P < 0.05$ ) marking significances compared to 2006.

show that rest time at 11 points increased significantly from 2006 to 2010 by 38.4% ( $z = -4.706$ ,  $p = 0.000$ ,  $r = 0.43$ ), 2011 by 35.2% ( $z = -4.192$ ,  $p = 0.000$ ,  $r = 0.38$ ), 2013 by 32.8% ( $z = -3.650$ ,  $p = 0.000$ ,  $r = 0.33$ ), 2014 by 23.1% ( $z = -3.545$ ,  $p = 0.000$ ,  $r = 0.32$ ), 2015 by 39.1% ( $z = -4.165$ ,  $p = 0.000$ ,  $r = 0.38$ ) and 2017 by 44.2% ( $z = -4.000$ ,  $p = 0.000$ ,  $r = 0.36$ ), see [Figure 3](#).

The Kruskal – Wallis test shows that the year influenced the rest time between games (Chi-square (8) = 30.683,  $p = 0.000$ ). Subsequent post-hoc tests (Dunn – Bonferroni tests)

show that rest time between games increases significantly from 2006 to 2010 by 60.1% ( $z=-3.450$ ,  $p=0.000$ ,  $r=0.41$ ), 2011 by 66.2% ( $z=-3.357$ ,  $p=0.000$ ,  $r=0.40$ ), 2013 by 69.0% ( $z=-3.668$ ,  $p=0.000$ ,  $r=0.44$ ), 2014 by 45.4% ( $z=-2.679$ ,  $p=0.007$ ,  $r=0.32$ ) and 2017 by 74.9% ( $z=-4.213$ ,  $p=0.000$ ,  $r=0.50$ ). Additionally, rest time between games increased from 2007 to 2010 ( $z=-2.606$ ,  $p=0.009$ ,  $r=0.31$ ), 2011 ( $z=-2.500$ ,  $p=0.012$ ,  $r=0.30$ ), 2013 ( $z=-2.761$ ,  $p=0.006$ ,  $r=0.33$ ), 2014 ( $z=-1.980$ ,  $p=0.048$ ,  $r=0.23$ ) and 2017 ( $z=-3.287$ ,  $p=0.001$ ,  $r=0.39$ ). Furthermore, there was also an increase in rest time between games from 2009 to 2017 ( $z=-2.250$ ,  $p=0.024$ ,  $r=0.27$ ) (Figure 3).

#### 4. Discussion

This study aimed to investigate game characteristics in MS over a decade (2006–2017) under the current scoring system to provide insights into hypothesised changes that may be of greater value for an ongoing discussion about further changing the scoring system. We hypothesised that an increase in game duration, longer rally time and rest time, increased rest time at point 11, and rest time between games can be observed over the decade. Our findings show the year's impact on all variables except for the points played. We found a significant increase for all the timing variables from 2006 to 2017.

Game duration increased overall by 54.0% from 2006 to 2017 and was also pronounced between the reference years of 2006 and 2013, 2014 and 2017. Abián et al. (2014) found a game duration increase of 12.1% for a 4-year interval from 2008 to 2012. However, Torres-Luque et al. (2020) found no significant increase in the total match duration compared with Olympic Games 2012 and 2016, but in the third game. Our game duration in 2017 ( $1474.9 \pm 272.8$  s) and the game duration reported by Abián et al. (2014) with  $1124.6 \pm 229.9$  s (2008 Beijing) or  $1260.3 \pm 267.1$  s (2012 London) were longer than those reported by Cabello Manrique (2003) with the old (the best-of-three games of 15 points for doubles and MS and 11 points for WS) scoring system ( $844.5 \pm 156.4$  s) and Chen and Chen (2008) with the current scoring system ( $803.4 \pm 34.7$  s). Therefore, our findings indicate that the change in the scoring system in 2006 did not affect the reduction in game length any longer in MS, conversely, there seems to be a steady increase if we look at the annual development over a decade for the first time.

Rally time was increased by 62.2% from 2006 ( $7.4 \pm 1.6$  s) to 2017 ( $12.0 \pm 2.8$ s), with additional significant differences from 2006 to 2011, 2013 and 2014. Additionally, rally time increased by 33% from 2015 and 2017. Our findings are consistent with studies of Abián et al. (2014), Laffaye et al. (2015) and Iizuka et al. (2017), which found changes in the rally length. However, Torres-Luque et al. (2020) did not demonstrate this over 4 years. Our mean rally time in 2017 was higher than those reported by others for the old (Cabello Manrique and González-Badillo, 2003:  $6.4 \pm 1.3$  s; Chen & Chen, 2008:  $8.1 \pm 6.3$  s) and current (Abian-Vicen et al., 2013:  $9.0 \pm .9$ s; Abián et al., 2014:  $10.4 \pm 2.1$  s and; Gawin et al., 2015:  $9.3 \pm 1.5$  s, among others) scoring systems.

Although the rally time increased continuously over the study period, shorter rallies ( $9.0 \pm 1.9$  s) were reobserved at the World Championship in 2015, and this could be because of a higher game speed (shots per rally:  $1.184 \pm 0.03$ ), temperature circumstances, less balanced games, or a higher risk on the part of players. Therefore, qualitative observations of this tournament match can provide further clarification.

While our results strengthen previous research findings (Abián et al., 2014; Iizuka et al., 2017; Laffaye et al., 2015), they extend them by pointing out the near-annual continuous increase in rally length covering a whole decade under the rally point system for the first time.

Rest time was increased by 49.3 % from 2006 ( $20.3 \pm 1.5$  s) to 2017 ( $30.3 \pm 4.1$  s), with significant differences observed from 2006 to 2010, 2013, 2014 and 2015. Laffaye et al. (2015), comparing the Olympic games between 1992 and 2012, and Iizuka et al. (2017), comparing MS and WS events of the 2007 and 2015 Japan Open Superseries, both reported a longer rest time. However, Abián et al. (2014) found no increase between 2008 and 2012. While Laffaye et al. (2015) and Iizuka et al. (2017) showed an increase in rest time when comparing the old and current scoring system, our data, as opposed to Abián et al. (2014), also could show this increase within the current rally-point-scoring system.

The increase in the rally and rest time effects increased the game duration over the observed period.

Additionally, rest time at point 11 increased from 2006 ( $56.8 \pm 11.7$  s) to 2017 ( $81.9 \pm 27.8$  s) by 44%. This was also reported by Abián et al. (2014) over 4 years (2008–2012) with a rest time at point 11 of  $79.6 \pm 11.6$  s in 2012. Therefore, it should be noted that these parameters exceeded the permitted range, considering the BWF's laws of badminton, which state that intervals during each game should not exceed 60 s when the leading score reaches 11 points (Badminton World Federation, 2023b). Interestingly, this value exceeded the permissible value of 60 s in 2007, only 1 year after the introduction of the rally-point scoring system; it rose far above the allowed value until 2017.

Moreover, rest between games was increased by 74.9% from 2006 ( $84.8 \pm 15.4$ ) to 2017 ( $148.3 \pm 14.7$ ). Abián et al. (2014) showed similar values and increased rest time between games of  $145.2 \pm 8.8$  s for 2012. It should also be noted that this parameter exceeded the allowed range. According to the BWF's laws of badminton, intervals may not exceed 120 s between the first and second games and between the second and third games (Badminton World Federation, 2023b). Our data show that the rest time between games increases continuously. In the first 4 years (2006–2009), the value was still below the permitted number of seconds; it passed the set range for the first time in 2010 and continuously increased until 2013. A prompt decrease was found for 2014 and 2015, possibly because of the referees' stricter interpretation of the rules. However, the value increased far above the permitted value in 2017. The rest time at point 11 and the rest time between games also affected the increase in game duration.

Furthermore, the intended effects of changing the scoring system more than 15 years ago appear to have faded due to an increase in all timing factors (rally time, rest time, rest at point 11 and rest between games) accumulated in the game and match durations. Therefore, stricter rule enforcement concerning the point 11 rest and rest between the games by the umpire, e.g. consistent use of cards (yellow and red) sanctioning players exceeding the permitted rest time, the introduction of an exact time limit for the rest time between the rallies, which could be similar to the shot clock implemented in tennis with a maximum of 25 seconds after the previous rally for the next service execution or a more basic change of the scoring system to the currently discussed best-of-five games of up to 11 points appears to be necessary to allow a reduction of the game duration/match duration.

When interpreting our results, it is important to take into account, that despite careful considerations, this study is subject to limitations. In science, prospective

trials should be used rather than retrospective analyses, which is not feasible when investigating professional badminton at the highest international levels of elite competition, especially when analysing long-term data over 10 years. Also, our study focused exclusively on MS; therefore, the illustrated development does not have to be similar to the other four events (WS, MD, WD, and MX). Moreover, this study is constrained by the limited number of matches and games, which are unevenly distributed over different years (4–7 matches and 9–17 games) and do not cover all tournament matches. Furthermore, small sample sizes may limit statistical power. This choice was made for two main reasons as follows: (1) the difficulty in retrieving videos of all matches during the World Championships from 2006 to 2017; older World Championships were not systematically stored and were unavailable for analysis, and (2) the rules of qualification for World Championships are based on the quota of players per country. Therefore, a reasonable difference exists between the leading and weaker countries, and several matches can become unbalanced in the first round. Another noteworthy limitation was partly poor video quality especially of the older matches, leaving possible misinterpretations due to unsuitable camera angles, reviews and cuttings; in some cases, rallies had to be excluded from the analysis. Consequently, counting the rest time at point 11 in six games was difficult, resulting in  $n = 121$  instead of  $n = 127$  for this factor. Notably, this affected the playing time; therefore, we could not work with real-time played. Although having to deal with limited video quantity and quality evaluation that both may have an effect on, at least in parts, research accuracy and data integrity, to the best of our knowledge, this present study used the only videos available that allowed studying such a long-time development. Moreover, using only one analyst may be considered a limitation with respect to data reliability, however, category definitions had previously been well-established.

For further studies, the video material should be more standardised, easier to access, and of higher quality including a video quality evaluation. Ideally, developments in all five disciplines should be analysed over a longer period, even suggesting the analysis of more matches per discipline (e.g. World Championships including all disciplines with a higher number of matches).

In conclusion, our findings allow us to conclude that the increase in all timing factors (rally time, rest time, rest at point 11, rest between games and game duration) accumulates in a longer match duration. However, the expected effects of changing the scoring system over the past 15 years appear to have faded. Additionally, our data gives a further indication that badminton is evolving towards longer rallies with longer rest times, including rest at point 11 and rest between games, exceeding the limits of the badminton regulation. Arguably, our findings may serve as a new attempt to change the scoring system to improve the sport's attractiveness and shorten match duration. Moreover, implementing strict rule enforcement during the rest at point 11 and rest between games could be an option to reduce the match duration. Therefore, the differences observed in the timing factors of badminton games could be important for the coaches and players in understanding the sport and managing different types of workouts and performance control and for tournament officials in creating competition schedules more specifically adapted to the characteristics of current badminton.

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